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Post Frac Permeability and Drainage Area Estimation in Tight Gas Reservoirs Through Type Curves for Optimization of Economic Model

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Abstract

Exploiting unconventional resources, such as shale gas, tight gas, and oil, has become imperative for the sustainability of the E&P industry due to the rapid decline of conventional resources. Hydraulic fracturing is a common practice for creating pathways in such reservoirs to facilitate flow. However, estimating post-frac reservoir parameters, such as permeability and drainage area, traditionally involves pressure build-up surveys with extended shut-in time, leading to production deferment. This paper discusses the type curves for determining these parameters without loss of production, furthermore, optimizing the economic model for execution of new fracs in the same field.

In this paper, production data of well X-2 is analyzed using Blasingame, Agarwal-Gardner, and Log-log type curves. The Blasingame type curve accounts for material balance time instead of actual producing time, thereby causing the boundary dominated to fall on analytical harmonic stem unlike Fetkovich type curve. The Agarwal-Gardner method plots inverse pressure derivative type curves which causes easier identification of different flow regimes compared to Blasingame. Apart from this conventional well testing log-log type curve is also used with modifications pertaining to pressure & removal of changing rates.

The matched production data using different type curves had computed almost similar range of permeability and drainage radius. Based on these results, the previous economic model for the field was updated, providing an opportunity to create feasibility for further development of the field with additional potential fracs. The sensitivities performed resulted in an optimum net present value (NPV). Adopting the type curve methodology would also reduce frac well's long production shut-in times, significantly saving annual surveillance expenditure ranging from $50-70k per well. The overall approach would optimize the field profitability, reduce production downtimes, and help to optimize the field development strategy.

Estimation of post frac parameters using various type curves does not incur production loss in terms of extended shut-in time with steady execution of development projects for sustainability of production & reserves.

Introduction

The global demand for energy has prompted an increased focus on unconventional gas resources, with tight gas reservoirs emerging as a significant contributor to the global natural gas supply. Characterized by low permeability and complex geological structures, these reservoirs present unique challenges in estimating key parameters such as permeability and drainage area. Accurate determination of these parameters is crucial for optimizing production strategies and enhancing the economic viability of tight gas reservoir projects [1].

In this context, the use of type curves has gained considerable attention as a powerful analytical tool for evaluating reservoir properties and forecasting production behavior. Type curves provide a graphical representation of well performance under various reservoir conditions, facilitating the estimation of key parameters crucial to reservoir characterization and management [2]. However, the application of type curves in the estimation of permeability and drainage area in tight gas reservoirs requires a comprehensive understanding of the interplay between reservoir characteristics, fluid dynamics, and production mechanisms.

This case study aims to contribute to the evolving body of knowledge surrounding the estimation of permeability and drainage area in tight gas reservoirs through the application of advanced type curve analysis techniques. By integrating theoretical models with empirical data, this study seeks to provide insights into the fundamental principles governing the flow of gas in these complex reservoirs, thus enabling more accurate predictions of reservoir performance and facilitating informed decision-making in the development and exploitation of tight gas resources.

Geological Challenges

Tight gas reservoirs present a complex geological landscape, often exhibiting heterogeneous and compartmentalized features that make the identification of sweet spots particularly challenging. Typically, the use of multi-azimuth seismic surveys is employed to locate these elusive sweet spots. Further complicating matters are the distinct petrophysical attributes observed in these reservoirs, including aspects such as porosity, phase trapping, permeability jail, capillary pressure, and initial displacement pressure. These properties play a pivotal role in differentiating tight gas reservoirs from their conventional counterparts. Below, we delve into an in-depth discussion of each of these defining properties.

The Challenge of Geology

The geological complexity of tight gas reservoirs stems from a range of post-depositional diagenetic processes, encompassing phenomena such as mechanical compaction, grain replacement, mineral dissolution, and the cementation of minerals like quartz. These processes induce permanent alterations in the initial pore structure, leading to increased tortuosity and a rise in the number of isolated or disconnected pores. Consequently, these reservoirs exhibit a notably reduced storage capacity. Moreover, the productivity of the reservoir is hindered as diagenesis processes block a significant portion of the pore throats, often resulting in an in-situ permeability. Such challenges significantly impede the identification and drilling of sweet spots. Employing advanced technologies, such as multi-azimuth seismic surveys, can prove invaluable in this context.

Porosity in TGRs

Given the intense overburden that tight gas reservoirs are subject to, the measurement of porosity must account for the impact of applied stress. The variations in porosity at different stress levels are crucial in comprehending how the reservoir will behave as gas is produced and reservoir pressure declines. This stress behavior is especially pertinent in the analysis of hydraulic fracturing performance.

Understanding Permeability Challenges

Similar to porosity, the permeability of tight gas reservoirs is also subject to the effects of stress. Stressed permeability is consistently lower than air permeability. This is primarily due to the low permeability of sands with small pore throats. As the net overburden increases, the permeability experiences a significant reduction. Relying on air permeability to estimate the productivity index can lead to an overestimation of reserves. Hence, stressed permeability is preferred in estimating practical reserves.

Challenges of Phase Trapping and Permeability Jail

The intricate pore structures found in tight gas reservoirs, dominated by slot-shaped and sheet-like pore throats, combined with the high effective overburden, give rise to specific challenges concerning multiphase flow. This intricate structure can lead to gas production impediments, notably through phenomena such as phase trapping. In the micro pore structure system inherent to tight gas reservoirs, a slight increase in water saturation can result in the blockage of the majority of pore throats, leading to a decrease in effective gas permeability. This phenomenon is known as phase trapping. Additionally, the high effective stresses within tight gas reservoirs may lead to a significant decrease in relative permeabilities for both gas and water.

Type Curves

Type curves are the unconventional & graphical method to estimate the well and reservoir parameters and it is based on the numerical simulation in which the actual well production data are analyzed and compared. Decline curve analysis method developed by the Arps in 1945 and it has been the most common method for reserves estimation and production forecasting due to its reliability and simplicity. In recent years as the conventional resources are declining and the industry shifted to exploring more and more tight gas resources and the production forecasting and reserves estimation for tight gas wells are problematic using the traditional curve method. The reason is that Arps' method is only worked when the boundary dominated flow achieved. In unconventional reservoirs, such decline is not achieved in tight wells during early production life where the early forecasting and financial decisions are important. To overcome this issue Fetkovich combines the transient radial flow solutions to the Arps' boundary dominated flow equation to develop the log-log production decline type curve. Since the end of the transient radial flow and start of boundary dominated flow can vary based on the distance to boundary, Fetkovich applies dimensionless variables (rate and time) to breakdown all solutions at a single end of radial flow time [3]

In recent years, the development of innovative type curves has revolutionized the process of determining post-frac reservoir parameters [4]. These type curves serve as a powerful analytical tool that facilitates the estimation of critical parameters such as permeability and drainage area without the need for prolonged shut-in periods. By leveraging advanced mathematical models and data analytics, these type curves enable engineers to make accurate predictions and assessments in a more time-efficient manner. This not only minimizes production deferment but also optimizes the overall operational efficiency of the extraction process.

Three different methods are used to estimate permeability, drainage area and Gas Initial In-place (GIIP) for low permeable and tight gas reservoirs.

- Palacio Blasingame
- Agrwal Gardner
- Log-log type curve

Blasingame

Blasingame devised a production decline method that addresses these issues, utilizing a form of superposition time function that necessitates only one depletion stem, known as the harmonic stem, for type curve matching. One notable advantage of this approach lies in the use of type curves for matching that are akin to those employed in Fetkovich's decline analysis, yet without the need for empirical depletion stems. When plotted using Blasingame's superposition time function, the analytical exponential stem of the Fetkovich type curve transforms into a harmonic one. This transformation is particularly significant considering that plotting the inverse of flowing pressure against time results in a harmonic decline during pseudo-steady state depletion at a constant flow rate.

Blasingame's method essentially enables depletion at a constant pressure to simulate depletion at a constant flow rate. Notably, Blasingame and colleagues demonstrated that boundary-dominated flow, characterized by both declining rates and pressures, manifests as pseudo-steady state depletion at a constant rate, provided that the rate and pressure decline monotonically.

The enhancements made by Blasingame to the Fetkovich style of production decline analysis involve the introduction of two additional type curves, the "rate integral" and "rate integral derivative," which aid in achieving a more distinctive match (Figure 1). These advancements build upon Fetkovich's approach, which combined the analytical solution for transient flow of a single-phase fluid at a constant wellbore flowing pressure with the empirical Arps equations for boundary-dominated flow.

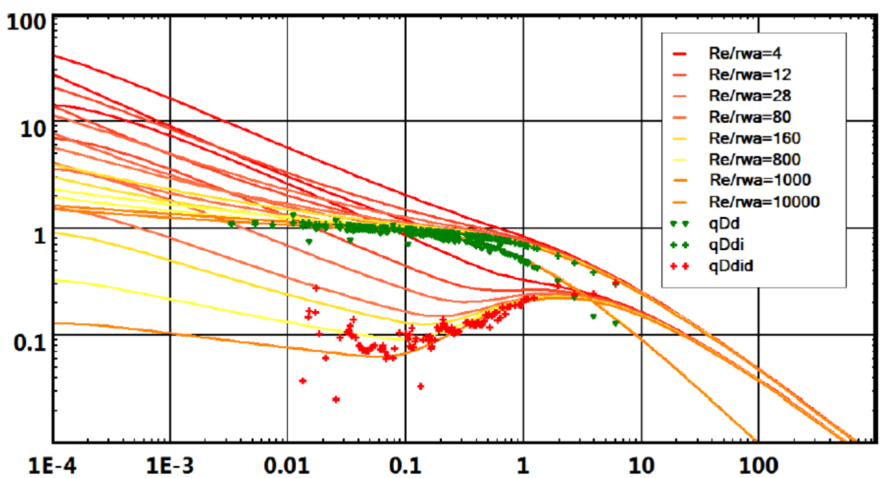


Figure 1—Blasingame Type Curve

Type curve developed by Rate Function with Analytical Transient Stem and Analytical Harmonic Decline Stem.

Calculate Normalize Rate Integral: $qi = \frac{1}{tC}\int_0^{tC} \frac{q}{\Delta p}\, dt$

Calculate Normalizes Rate Integral Derivative: $qid = tc\frac{dql}{dtc}$

Agrwal Gardner

Agarwal and Gardner presented a fresh set of decline type curves aimed at facilitating the analysis of production data. Their approach was build upon the methodologies proposed by both Fetkovich and Palacio-Blasingame, incorporating the notion of equivalence between solutions for constant rate and constant pressure. Additionally, their method includes primary and semi-log pressure derivative plots presented in an inverse format for decline analysis.

To account for changing production conditions, Agarwal et al. incorporate variations in flowing pressure and production rate, normalizing the production data using material balance time. Changes in gas properties are accommodated using gas pseudo-time. They propose the utilization of rate-cumulative type curves for estimating gas or oil in place, where dimensionless cumulative production is plotted against dimensionless rate.

Agarwal-Gardner used the conventional well test dimensionless rate and dimensionless time definitions and introduce type curves for **Pressure Derivative** and **Semi Logarithmic** (well-test) **Derivative** (plotted as inverse) to estimate in-place volume.

- Inverse Pressure Derivative calculated by: $\frac{1}{DER} = \left[tc \frac{d\left(\frac{\Delta p}{q}\right)}{dtc} \right]^{-1}$

Log-Log Type Curve

Fetkovitch introduced a novel set of type curves that expanded the Arps type curves into the transient flow region. Acknowledging that decline curve analysis applied only during the boundary-dominated flow period, namely the depletion period, he recognized the limitation of conventional decline curve methods in analyzing the early production life of a well. By incorporating analytical flow equations to generate type curves for transient flow, he seamlessly integrated them with the Arps empirical decline curve equations. Thus, the Fetkovitch type curves were formulated to encompass both the early-time transient flow and the late-time boundary-dominated flow, ensuring continuity throughout the entire production lifecycle.

Under the condition of constant pressure production, a well adheres to one of these curves. Fetkovitch attributed the success of Fetkovitch type curves to the prevalent operation of most oil wells under wide-open conditions, maintaining the constant minimum achievable pressure. Additionally, he observed instances where the hyperbolic decline coefficient, b, as determined using the Arps decline curves, exceeded 1 (b typically falls within the range of 0 to 1). He proposed an explanation for this occurrence, suggesting that such data still resided in the transient flow regime and had not transitioned to the boundary-dominated regime. When attempting to fit transient flow data to a hyperbolic decline, values of b greater than 1 could consequently emerge.

From the period of transitional flow, it is possible to approximate the skin and permeability, whereas during the boundary-dominated phase, an estimation of the estimated ultimate recovery (EUR) becomes feasible. During the initial stages, the data typically aligns with a transient state, while during the latter stages, it conforms to a pseudo-steady state. However, despite these developments, the pursuit of obtaining estimates for EUR and future rates from transient data remains unrealized.

Field Description

Field Z

Field Z was discovered by drilling Z-1 in Lower Goru Basal Sands. DST was conducted which showed the Upper Basal Sand (UBS) reservoir to be tight since no flow observed. However Lower Basal Sand (LBS) reservoir produced gas at a rate of ~21 MMscfd against WHFP of 3425 psig during DST. Z-2 well was drilled and penetrated the LBS ~250′ updip to Z-1, therefore, it is planned to workover Z-2 to squeeze the currently perforated UBS and re-access LBS to optimally deplete LBS remaining reserves leftover by Z-1. The production history of the well is shown in figure.

Structural Mapping

Mapping of the Field Z structure was carried out on Pre-stack Time migrated version of 3D seismic data of Merge-2 by incorporating the drilling results of nearby wells, which helped to understand the structural complexity of the prospect and determine its areal extents. Regionally, the mapping area is structurally complex with numerous splay faults. Seismic data quality in the area is fair to good. All the available geological and performance data of the existing wells was integrated to evaluate the potential associated with this prospect and to delineate this deepening opportunity.

Structure

Z structure is an extension play type similar to proven structures other operating blocks. It is a tilted fault block forming a 3-way dip closure against the structure bounding fault. The closure to the east is provided by a north-south trending, down-to-the east main bounding fault having a throw of approx. 500-700 ft, while closure to the west, north and south is provided by dip. There is a west dipping splay fault that runs parallel – sub-parallel to main bounding fault. The closure of field Z structure can be mapped from Upper Basal Sand to the deepest possible reservoir level.

For Lower Basal Sand Reservoir, the crest of the structure is at -11,130 ft TVDSS and structural spill point is at -11,520 ft TVDSS. The areal closure up to SSP is ~ 1250 acres. For Sembar Formation, crest of the structure is at -12,640 ft TVDSS. Areal closure up to SSP at -13,000 ft TVDSS is ~ 1550 acres.

Reservoir

Prospectivity of Lower Basal Sand has already been proved in field Z structure. However, the new discovery of Field Y has significantly improved the prospectivity of Lower Basal Sands by discovering multiple hydrocarbon bearing sands. The discovery of Y field and its deeper sands are one of the key factors for deepening Z-2, which has drilled only ~470 ft of 1875 ft total gross thickness of Lower Basal Sand.

The UBS encountered in Field is 40 ft with ~9 % and average water saturations 40%, while LBS encountered in the field with ~100 ft net pay with average porosity ~12 % and average water saturations 25%.

Gas In place Volumes

GIIP for UBS Reservoir in Z Field are evaluated for the cases assuming structure is filled down from LKG to SSP (proved to possible category). Estimated volume is in the range of 7.5-16.4 Bscf assuming 7% average porosity, 40% average Sw and 45′ maximum net gas pay on the structure

LBS GIIP estimates with the structure filled down to individual Lobes′ LKG at 11,486′ TVDSS, 11,603′ TVDSS, 11,714′ TVDSS and 11,875 TVDSS respectively. The total GIIP is estimated to be 52.93 Bscf

Field Y

Y field is located in South Indus Basin and is characterized by gas accumulation in UBS & LBS formation. The field was discovered in 2013 through drilling of Y-1 well; the well encountered gas pay in UBS & LBS Reservoir Lobes. The UBS within this field was interpreted as tight gas potential reservoir.

It is a combination trap provided by both dip and faults. The closure to the east is provided by the main field bounding fault. Towards the north and west, closure is provided by a splay fault which dies out in the south where the closure is provided by the dip. UBS Gas Initial In-place (GIIP) estimation for are based on dynamic model results and are in the range of 53 Bscf for 1P and 86 Bscf for 2P Case.

Mapping Detail

Mapping of the Y Field was carried out on 3D processed Pre-Stack Time Migrated (PSTM) seismic data by incorporating Geological & Geophysical (G&G) data of Y Field wells. The mapping area is structurally complex with numerous splay faults. The seismic data quality in the mapping area is fair with better signal-to-noise (S/N) ratio.

Structural Configuration

The field is located on the up thrown side of a NW-SE oriented down to the East field bounding fault. The closure to the South is provided by dip, to the North and West, closure is provided by a combination of dip and fault, while the critical closure to the East is provided by the field bounding fault.

Reservoir properties

The UBS encountered in Field is tight i.e., permeability being less than 1 mD with average porosity ~12 % and average water saturations 40%, therefore, it requires hydraulic fracturing to establish producibility. Well Y-5 is drilled as a vertical well and produced ~0.45 MMscfd gas initially after fracturing. The reservoir thickness varies between 24 ft to 63 ft, and it is continuous throughout the field.

Pressure Transient Analysis

A PBU survey was conducted in 2016 to determine reservoir properties and the extent of the reservoir.

Early time region shows type curves as high wellbore storage affects has masked the ETR response i.e. fracture presence. Middle Time Region depicts IARF interpretation as reservoir to be very tight i.e. 0.034 md. While Late Time Region shows the derivative trend indicates possible constant pressure source which requires more shut-in for confirmation.

Hydraulic Fracturing

Hydraulic fracturing, commonly known as fracking, has emerged as a groundbreaking technology in the E&P industry, especially in the extraction of unconventional resources. This technique involves the creation of fractures in reservoir rocks, thereby enabling the release of previously inaccessible hydrocarbons.

Field Z

Frac Design & Execution. Frac design was optimized based on economic consideration and it was found that 150 ft half-length maximizes NPV. It relied on reservoir data from the DFIT analysis of Z-2, MDT results from Z-2X, and core samples from Y-3. Post frac estimated design rates at different fracture half-length mentioned in Figure 4.

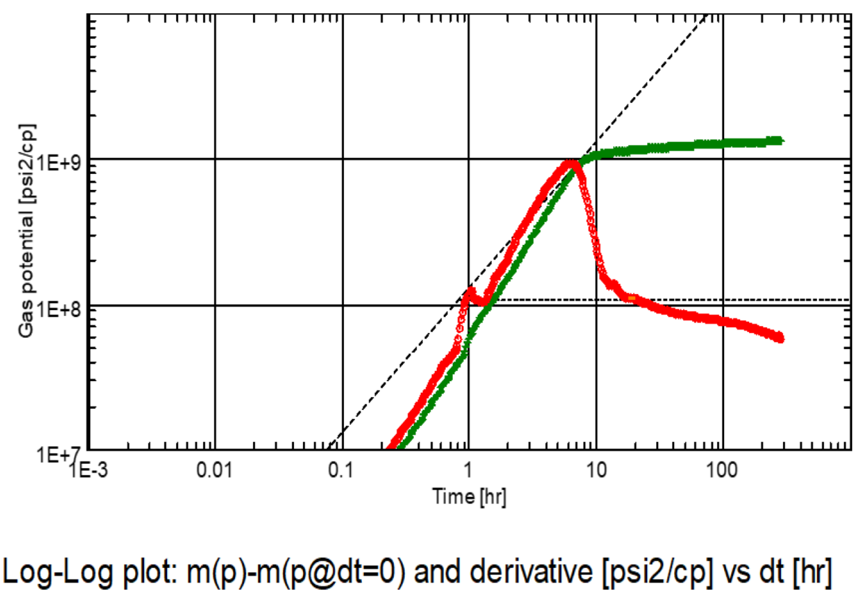


Figure 2—Log Log Plot

Name	Value	Unit
Main Model Parameters		
TMatch	0.625	[hr]-1
PMatch	4.64E-9	[psi2/cp]-1
C	0.0336	bbl/psi
k.h, total	2.05	md.ft
k, average	0.0348	md
Pi	5422.73	psia
Derived & Secondary Parameters		
Semilog Line (Naimat West-5 (PBU Survey) MHT # 410 BOT		
From	-278.477	hr
To	-277.177	hr
Slope	2.58184E+8	psi2/cp
Intercept	1.48457E+9	psi2/cp
M(p)@1hr	8.34936E+8	psi2/cp
PMatch	4.46E-9	[psi2/cp]-1
k.h	1.97	md.ft
k	0.0334	md
p*	5825.01	psia
Skin	0.523	--
Delta P Skin	339.377	psi

Figure 3—Log Log Data

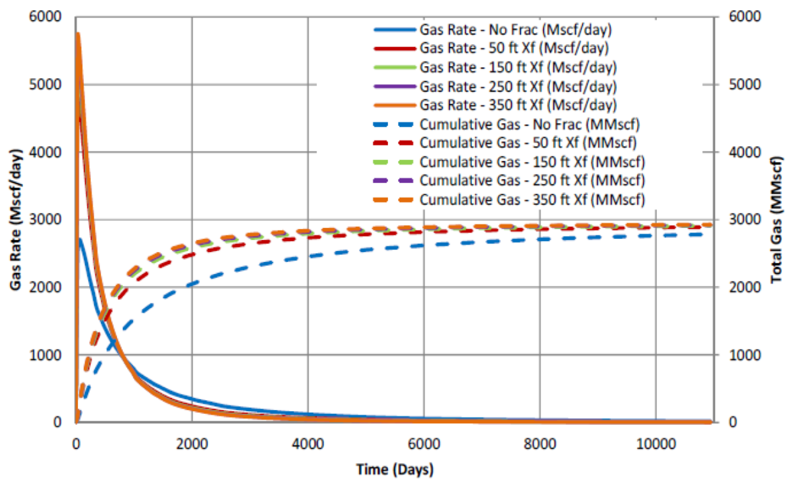


Figure 4—Post Frac Design Rates Sensitivities

The fracture treatment plan targets the Upper Basal Sand interval 11,004 and 11,050 ft MD with an estimated propped fracture half-length of 155 ft (based on a 0.50 lb/ft² threshold). It involves using 2,000 gallons of acid and 79,500 gallons of 40 lb linear/borate crosslinked fluid to carry 77,000 lbs of 20/40 mesh CarboProp proppant. Presence of minor splay fault concern addressed in conservative treatment plan.

Prior to main treatment, Diagnostic Fracture Injection test was performed, and analysis shows higher NWBF losses (~2000 psi) with closure gradient of 0.73 psi/ft & frac gradient of 1.16 psi/ft. During main treatment, started injection at 20 bpm rate and pump 50 bbls 10% HCl acid and two proppant slugs (0.25ppa & 0.5 ppa) in pad volume (~1100 bbls) to reduce NWBF (Figure 5). Total volume pumped was ~2000 bbls frac fluid with 76 klbs proppant of 30/50 size (max conc 4 ppa).

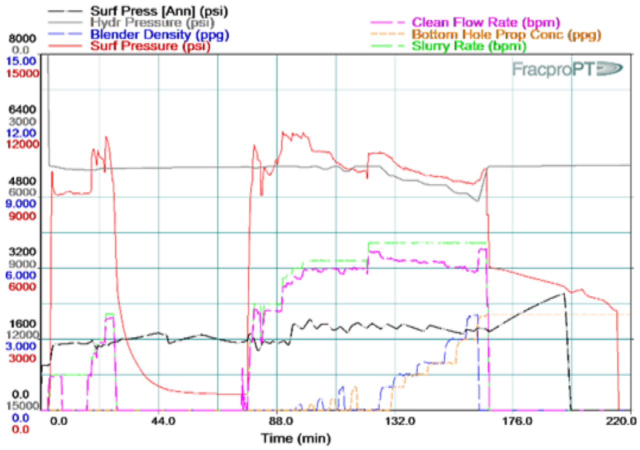


Figure 5—Treatment Plot of Well NX-2

Post Frac Performance. During flowback well took time to clean and production increases gradually to ~2.5 MMscd gas along with 230 BCPD and 720 BWPD. Post frac estimated design rates at different fracture half length.

Field Y

Frac Design & Execution. Frac design was optimized considering the max NPV and it was found that 200 ft half-length maximizes NPV (10% discount rate). It relied on data from the DFIT analysis of Y-5 and fracture treatment & production history from Z-2 offset, and core samples from Y-3.

The fracture treatment plan targets the Upper Basal Sand interval 10,627 and 10,637 ft MD with an estimated propped fracture half-length of 210 ft (based on a 0.50 lb/ft² threshold). It involves using 2,000 gallons of acid and 100,200 gallons of 40 lb linear/borate crosslinked fluid to carry 110,000 lbs of 30/40 mesh proppant.

DFIT was carried out which shows breakdown gradient of 1.6 psi/ft and closure gradient of 0.83 psi/ft. Prior to main treatment, mini frac was carried out and analyzed. Observed higher NWBF losses (~2000 psi) which was perforation dominated. During main treatment, started injection at 21 bpm rate and pump 50 bbls acid and two proppant slugs (0.25ppa & 0.5 ppa) in pad volume (~38%) to reduce NWBF (Figure 6). Total volume pumped was ~2168 bbls frac fluid with 110 klbs proppant of 30/50 size (max conc 4 ppa).

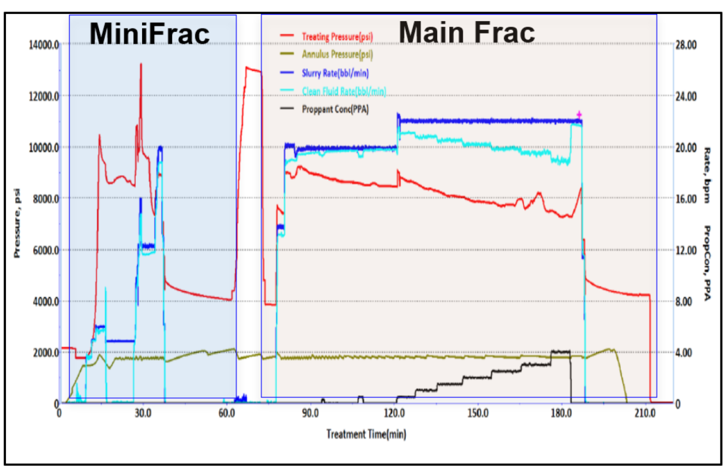


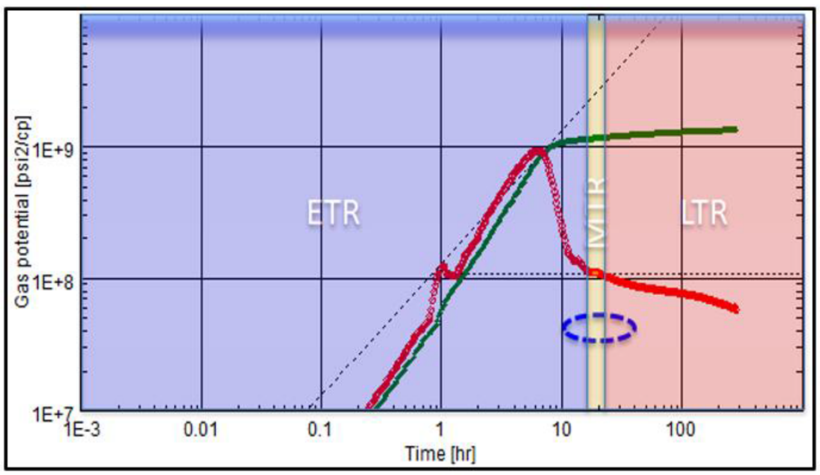
Figure 6—Treatment Plot of Well Y-5

Post Frac Performance. Post frac well offloaded ~1459 barrels (58%) frac fluid, while well was tested at 0.43 MMscfd gas and 20 BOPD. PBU was carried out with surface shut-in after 3 weeks of production, permeability estimate was 0.034 mD, positive skin of +0.5. Fracture response was not evident in PBU which could have been masked by ETR response. Post frac production was matched with permeability of 0.06 mD and a positive skin of +3. Later on perforations were made in the UBS interval at 10,637' – 10,657' KB but the well's behavior remained consistent. After 4 years of production static BHP was recorded at 4,945 psia (700 psi depletion).

Case Study

In this paper two distinct fields with different geological and petrophysical have been discussed to develop robustness of the type curves.

In the first step production data has been analyzed by using the Blasingame with normalized rate integral (Figure. 8) and integral derivative (Figure. 9) followed by the Agrwal with inverse derivative plot (Figure. 10).



Name	Value	Unit
Main Model Parameters		
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C	0.0336	bbl/psi
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k, average	0.0348	md
Pi	5422.73	psia
Derived & Secondary Parameters		
Semilog Line (Naimat West-5 (PBU Survey) MHT # 410 BOT		
From	-278.477	hr
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PMatch	4.46E-9	[psi2/cp]-1
k.h	1.97	md.ft
k	0.0334	md
p*	5825.01	psia
Skin	0.523	--
Delta P Skin	339.377	psi

Figure 7—PBU Interpretation

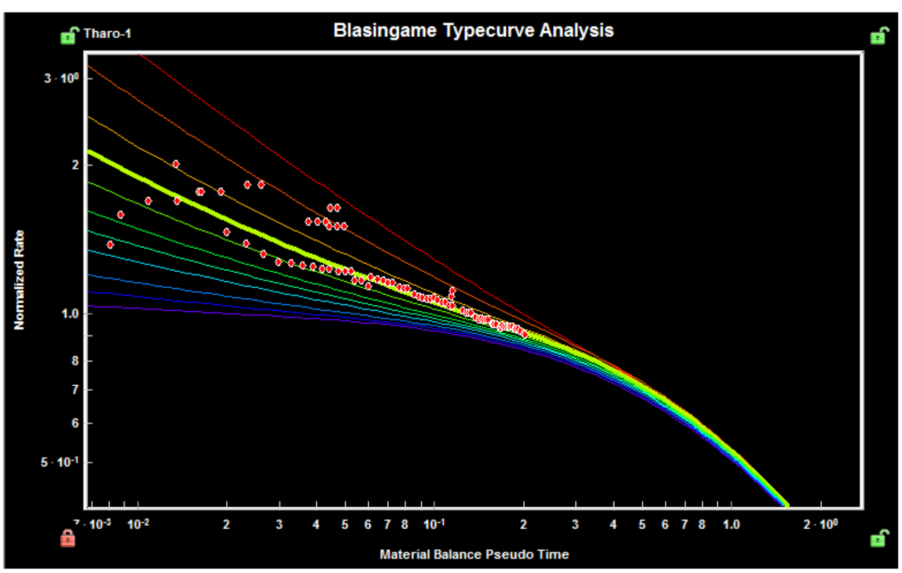

Figure 8—Normalized Rate Integral

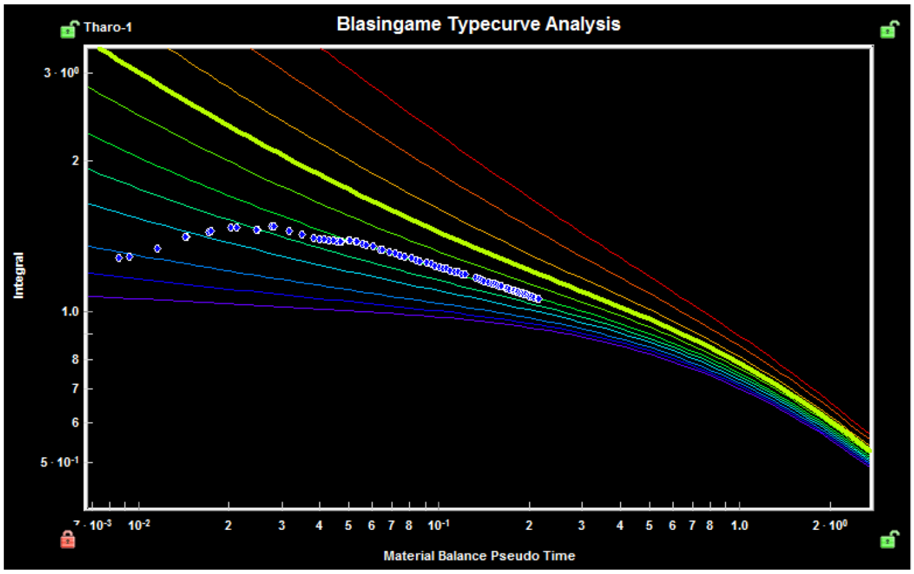

Figure 9—Rate Integral Derivative

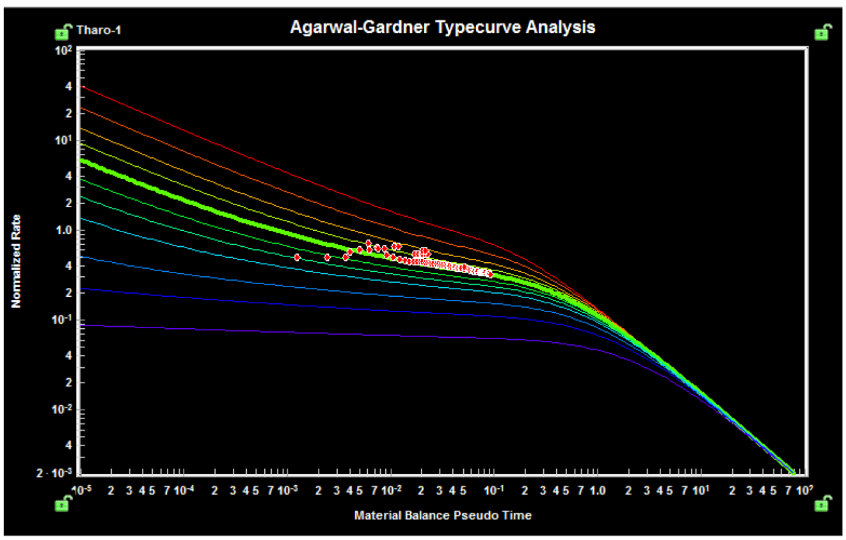

Figure 10—Agarwal Semi-Log Curve

Results/Conclusion

The economic viability of hydraulic fracturing operations remains a critical concern for the E&P industry. Balancing the costs of production with the market demand for energy resources is essential to ensure the long-term sustainability of such operations. While hydraulic fracturing has proven to be a crucial tool in unlocking the potential of unconventional resources, it also presents certain complexities, especially when it comes to accurately estimating post-frac reservoir parameters like permeability and drainage area. Traditionally, this estimation process required time-consuming pressure build-up surveys and extended shut-in periods, leading to substantial production deferment. To overcome this challenge, type curves serve as a powerful analytical tool that facilitates the estimation of these critical parameters without causing any significant loss of production. Moreover, the integration of these type curves into the decision-making process has significantly enhanced the industry's ability to streamline operations and maximize the output from unconventional reservoirs.

Following are the results of the analysis and key outcomes of the study.

- Types curves have computed almost similar range of permeability and drainage radius that calculated through pressure transient analysis.
- Production deferment would be reduced by avoiding long production shut-in times for frac wells.
- Annual surveillance expenditure savings ranging from $50-70k per well.
- Early field development plan can be worked out with steady execution of development projects for sustainability of production & reserves.
- Well placement at sweet spots and fracturing job would be optimized.

Nomenclature

BBL = Barrels
BSCF = Billion Standard Cubic Feet
BWPD = Barrels of Water Per Day
FT = Feet
FWHP = Flowing wellhead pressure
BHP = Bottom hole Pressure
mD = milli Darcy
MM = Million
MMSCFD = Million Cubic feet per day of gas at standard conditions
BCPD = Barrels of Condensate Per Day
NPV = Net Present Value
PSI = Pounds per square inch
q = Production rate at time *t*
tc = Producing days
Δp = Pressure drop, psi
dt = Time derivative
lb = Pounds
TVDSS = True Vertical Depth Subsea
SSP = Structural Spill Point

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